

SCH3U Unit Test 3: Stoichiometry

Formulas:

$$n = \frac{m}{M}$$

$$n = \frac{N}{N_A}$$

$$\text{Conversion factor} = \left(\frac{\text{want}}{\text{have}} \right)$$

$$\% \text{ mass} = \frac{m_{\text{part}}}{m_{\text{whole}}} \times 100\%$$

$$\% \text{yield} = \frac{\text{experimental yield}}{\text{theoretical yield}} \times 100\%$$

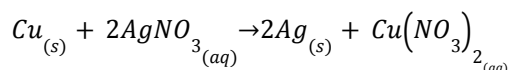
$$N_A = 6.022 \times 10^{23}$$

$$\text{Molecular Multiplier} = \left(\frac{m_{\text{molecular}}}{m_{\text{empirical}}} \right)$$

Part A: Multiple Choice

- What is the molar mass of NaCl?
 - 56.23 g/mol
 - 58.44 g/mol
 - 61.42 g/mol
 - 67.12 g/mol
- What is the molar mass of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$?
 - 197.18 g/mol
 - 159.61 g/mol
 - 203.56 g/mol
 - 249.71 g/mol
- How many moles of Silver are present in 6.78×10^{25} Ag atoms?
 - 113 mol
 - 117 mol
 - 6.022×10^{23} mol
 - 18.5 mol
- How many molecules are present in 4.5 moles of H_2O ?
 - 6.022×10^{23} molecules
 - 1.35×10^{24} molecules
 - 2.7×10^{24} molecules
 - 5.4×10^{24} molecules
- How many hydrogen atoms are present in 4.5 moles of H_2O ?
 - 6.022×10^{23} molecules
 - 1.35×10^{24} molecules
 - 2.7×10^{24} molecules
 - 5.4×10^{24} molecules

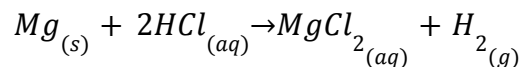
Use the following chemical equation to answer questions 6 – 10:



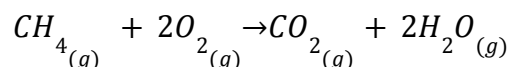
6. What is the molar mass of Ag?
 - a. 196.97 g/mol
 - b. 107.87 g/mol
 - c. 26.98 g/mol
 - d. 74.92 g/mol
7. What is the molar mass of $\text{Cu}(\text{NO}_3)_2$?
 - a. 93.55 g/mol
 - b. 125.55 g/mol
 - c. 165.87 g/mol
 - d. 187.56 g/mol
8. If 3.2 moles of Cu are used in the reaction, how many moles of Ag are produced?
 - a. 6.4 mol
 - b. 3.2 mol
 - c. 1.6 mol
 - d. 32 mol
9. If 7.68×10^{23} molecules of AgNO_3 are used in the reaction, how many molecules of $\text{Cu}(\text{NO}_3)_2$ are produced?
 - a. 1.54×10^{24} molecules
 - b. 7.68×10^{23} molecules
 - c. 3.84×10^{23} molecules
 - d. 1.28 mol
10. If 7.2 moles of Cu are mixed with 13.0 moles of AgNO_3 , what is the limiting reactant?
 - a. Cu
 - b. Ag
 - c. AgNO_3
 - d. $\text{Cu}(\text{NO}_3)_2$
11. If a reaction is expected to produce 12.5 g of H_2SO_4 , but only 12.1 g are collected. What is the percent yield?
 - a. 100%
 - b. 96.8%
 - c. 95%
 - d. 1.03%
12. If a reaction has a theoretical yield of 7.45 g of Na_2SO_4 , and a percent yield of 85%, how many grams of Na_2SO_4 are produced?
 - a. 633 g
 - b. 7.45 g
 - c. 8.56 g
 - d. 6.33 g
13. Which of the following chemicals would have the empirical formula CH_2O ?
 - a. $\text{C}_3\text{H}_6\text{O}$
 - b. $\text{C}_3\text{H}_6\text{O}_3$
 - c. CHO
 - d. $\text{C}_4\text{H}_8\text{O}_2$
14. Benzene has an empirical formula of CH , but a molar mass of 78.12 g/mol. What is the molecular formula of benzene?
 - a. C_5H_5
 - b. CH_4
 - c. 6CH
 - d. C_6H_6
15. What tool might we use to find the percent composition of a hydrocarbon fuel?
 - a. The Percentalyzer™
 - b. A balance and a calculator
 - c. CH combustion analyzer
 - d. A dehydration chamber
16. In the hydrate $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, what percent of the compound is made up of water?
 - a. 36 %
 - b. 20 %
 - c. 56 %
 - d. 500 %
17. If a compound contains 4.5 mol of C and 12.0 mol of H are present in a compound containing C and H, what is the empirical formula?
 - a. CH_3
 - b. C_3H_8
 - c. C_2H_6
 - d. CH
18. What is the percent composition of NaCl ?
 - a. 61 % Cl, 39 % Na
 - b. 39 % Cl, 61 % Na
 - c. 50 % Na, 50 % Cl
 - d. Impossible to determine for ionic compounds
19. What is the empirical formula of $\text{C}_6\text{H}_9\text{O}$?
 - a. $\text{C}_2\text{H}_3\text{O}$
 - b. CHO
 - c. $\text{C}_{12}\text{H}_{18}\text{O}_2$
 - d. $\text{C}_6\text{H}_9\text{O}$
20. What is the molecular formula of a compound with an empirical formula of $\text{C}_6\text{H}_9\text{O}$, and a molar mass of 97 g/mol?
 - a. $\text{C}_2\text{H}_3\text{O}$
 - b. CHO
 - c. $\text{C}_{12}\text{H}_{18}\text{O}_2$
 - d. $\text{C}_6\text{H}_9\text{O}$

Part B: Application Questions

1. If 5.85 g of HCl are consumed in the following reaction, how many moles of MgCl_2 are expected? [3]

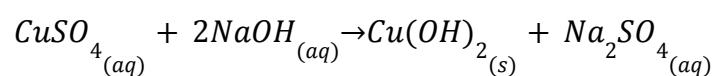


2. If 4.50×10^{23} molecules of CH_4 are burned, in sufficient oxygen, how many grams of H_2O are expected? [4]



3. Determine the percent composition of CuSO_4 . [3]

4. If 5.40 g of CuSO_4 are reacted with 4.85 g of NaOH



a. Which reactant will be limiting? [1] _____

b. How many grams of Na_2SO_4 are expected? [3] _____

- c. If the reaction has a 90.0% yield, how many grams of Na_2SO_4 are produced? [2]
5. A compound contains C, H and O. The compound has a percent composition of 40.00 % C, 6.71 % H and 53.29 % O.
- a. Determine the empirical formula [3]
- b. If the compound has a molar mass of 150.15 g/mol, what is the molecular formula? [2]
6. A 1.50 g sample of hydrocarbon, containing C and H, undergoes complete combustion to produce 4.40 g of CO_2 and 2.70 g of H_2O . Analysis with a mass spectrometer indicates a molar mass of approximately 30 g/mol. What is the molecular formula of the compound? [6]

BONUS: In question 4, how many grams of the excess reactant will remain after the reaction is completed? (theoretical, not based on percent yield) [1]
